Signal Processing

Jnanesh Sathisha karmar - EE25BTECH11029

March 25,2026

Question

1.2.29 Consider the discrete-time signal $x(n) = (1/3)^n u(n)$, where $u(n) = 1$ for $n \geq 0$, and define $y(n) = x(-n)$. Then

$$\sum_{n=-\infty}^{\infty} y(n)$$

equals: (IN 2007)

a) $-2/3$ b) $2/3$ c) $3/2$ d) 3

Theoretical Solution

Method 1

First, let us explicitly write the expression for $y(n)$:

$$y(n) = x(-n) = \left(\frac{1}{3}\right)^{-n} u(-n) \tag{1}$$

Because $u(-n) = 1$ only for $n \leq 0$, the signal $y(n)$ is zero for all positive values of n. We can set up the required summation:

$$S = \sum_{n=-\infty}^{\infty} y(n) = \sum_{n=-\infty}^{0} \left(\frac{1}{3}\right)^{-n} \tag{2}$$

To evaluate this, we perform a change of variables. Let $k = -n$. As n goes from $-\infty$ to 0, k goes from ∞ down to 0.

$$S = \sum_{k=0}^{\infty} \left(\frac{1}{3}\right)^{k} \tag{3}$$

Theoretical Solution

This is a standard infinite geometric series of the form $\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}$, where the first term $a = 1$ and the common ratio $r = -\frac{1}{2}$. Since $|r| < 1$, the series converges.

$$S = \frac{1}{1 - \left(\frac{1}{3}\right)} = \frac{1}{1 - \frac{1}{3}} = \frac{1}{\frac{2}{3}} = \frac{3}{2} \tag{4}$$

This matches option **(c)**.

Theoretical Solution

A more elegant approach utilizes the properties of the Z-transform. The sum of a discrete-time sequence from $-\infty$ to ∞ is precisely its Z-transform evaluated at $z = 1$ (which corresponds to the Discrete-Time Fourier Transform evaluated at $\omega = 0$).

1. Derivation of the Z-transform of $x(n)$:

The two-sided Z-transform of a discrete-time signal $x(n)$ is defined fundamentally as:

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} \tag{5}$$

Substitute our specific signal $x(n) = \left(\frac{1}{3}\right)^n u(n)$ into the definition. Because the unit step function $u(n) = 0$ for all $n < 0$, the lower limit of our summation changes from $-\infty$ to 0:

$$X(z) = \sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^n z^{-n} \tag{6}$$

Theoretical Solution

By exponent rules, we can group the terms raised to the power of n together:

$$X(z) = \sum_{n=0}^{\infty} \left(\frac{1}{3}z^{-1}\right)^n \tag{7}$$

This is a standard infinite geometric series of the form $\sum_{n=0}^{\infty} r^n$. We know from calculus that this series converges to $\frac{1}{1-r}$ if and only if the absolute value of the common ratio is strictly less than 1 ($|r| < 1$).
Applying this formula where our common ratio is $r = \frac{1}{3}z^{-1}$:

$$X(z) = \frac{1}{1 - \left(\frac{1}{3}z^{-1}\right)} = \frac{1}{1 - \frac{1}{3}z^{-1}} \tag{8}$$

Defining the Region of Convergence (ROC): The sum only converges (and thus the Z-transform only exists) when $|r| < 1$. Let's solve for z:

$$\left|\frac{1}{3}z^{-1}\right| < 1 \implies \frac{1}{3}|z|^{-1} < 1 \implies |z| > \frac{1}{3} \tag{9}$$

Theoretical Solution

Therefore, the Z-transform is $\frac{1}{1-\frac{1}{3}z^{-1}}$ with an ROC of $|z| > \frac{1}{3}$. This ROC is critical because it tells us that evaluating the function at $z = 1$ (which we do in step 3) is mathematically valid, since $1 > 0.33$.

2. Time-Reversal Property:

If $y(n) = x(-n)$, its Z-transform is $Y(z) = X(z^{-1})$.

$$Y(z) = \frac{1}{1 - \frac{1}{3}(z^{-1})^{-1}} = \frac{1}{1 - \frac{1}{3}z} \tag{10}$$

The Region of Convergence (ROC) also inverts: $|z^{-1}| > \frac{1}{3} \implies |z| < 3$.

3. Evaluate at $z = 1$:

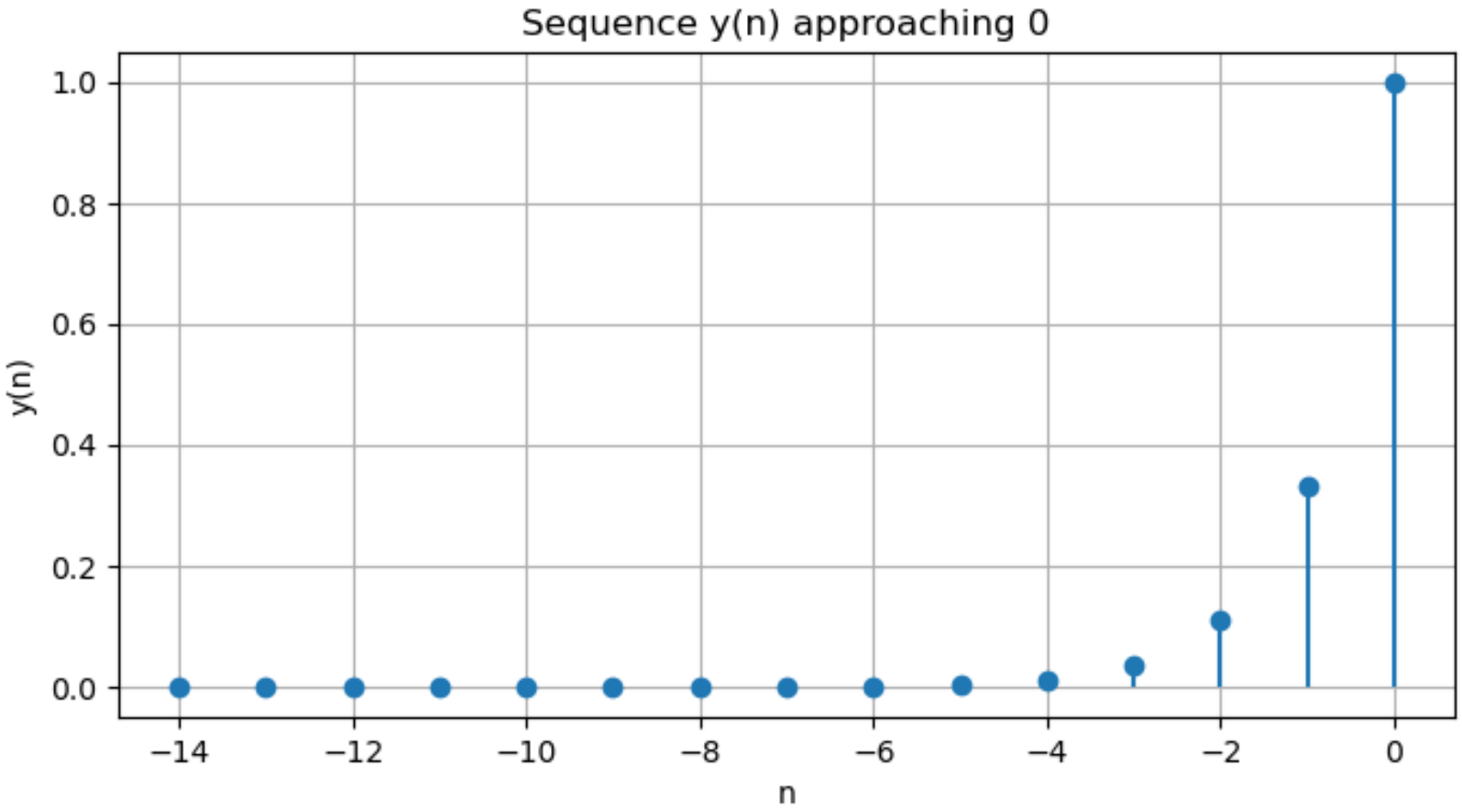
Because $z = 1$ lies safely inside the ROC ($1 < 3$), the infinite sum converges to exactly $Y(1)$.

$$\sum_{n=-\infty}^{\infty} y(n) = Y(1) = \frac{1}{1 - \frac{1}{3}(1)} = \frac{3}{2} \tag{11}$$

Python Code

```python
import numpy as np
import matplotlib.pyplot as plt
n = np.arange(-100, 1)
y = (1/3)**(-n)
numerical_sum = np.sum(y)
print(f"Theoretical Sum: 3/2 = {3/2:.5f}")
print(f"Numerical Sum (first 100 terms): {numerical_sum:.5f}")
plt.figure(figsize=(8, 4))
plt.stem(n[-15:], y[-15:], basefmt=" ")
plt.title('Sequence y(n) approaching 0')
plt.xlabel('n')
plt.ylabel('y(n)')
plt.grid(True)
plt.show()
```

Plot-Using only Python



```
└─ python3 main.py
Theoretical Sum: 3/2 = 1.50000
Numerical Sum (first 100 terms): 1.49794
```

Figure: n=5

```
└─ python3 main.py
Theoretical Sum: 3/2 = 1.50000
Numerical Sum (first 100 terms): 1.49999
```

Figure: n=10

```
└─ python3 main.py
Theoretical Sum: 3/2 = 1.50000
Numerical Sum (first 100 terms): 1.50000
```

Figure: n=100